

FOSSIL VERTEBRATES

Fossilized vertebrate remains are not as common as those of invertebrates, since many vertebrates lived on land, where fewer fossils form, and they evolved much later than the invertebrates. Fish were the earliest vertebrates to evolve, some dating back to Cambrian times. Their rapid evolution in the Silurian and Devonian led to amphibians, which first appeared in the Devonian Period. Dinosaurs flourished during the Mesozoic Era, towards the end of which mammals began to diversify.



EARLY FISH-LIKE VERTEBRATE

Loganellia sp.
A primitive, jawless, flattened "fish", *Loganellia* was covered with tooth-like scales. Up to 12 cm (4¾ in) long, it is found in Devonian rocks.



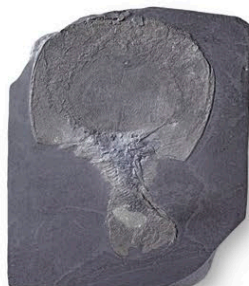
LOBE-FINNED FISH

Eusthenopteron foordi
The bones in the heavily built fins of this Late Devonian fish were similar to those in the limbs of land-dwelling vertebrates (tetrapods).



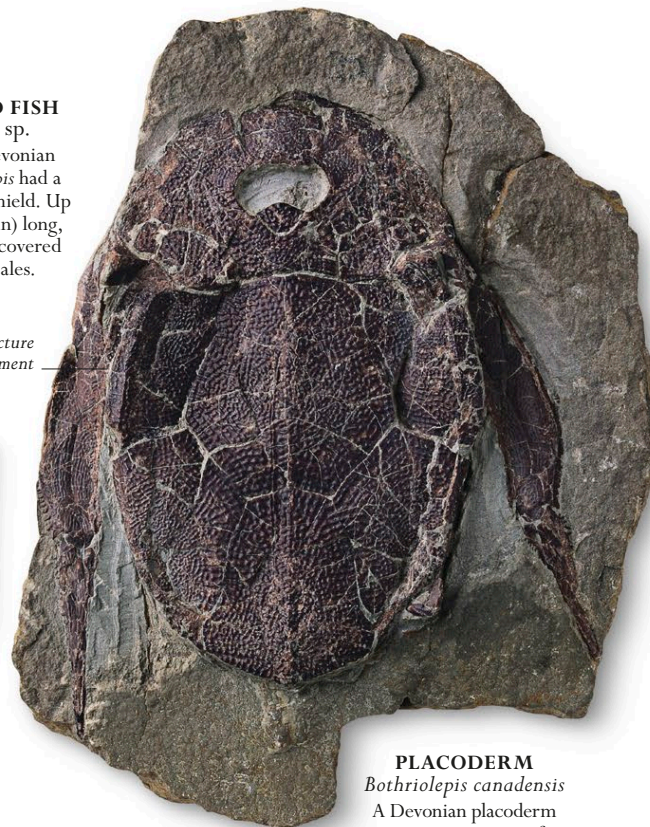
ZENASPID FISH

Zenaspis sp.
Found in Devonian strata, *Zenaspis* had a massive headshield. Up to 25 cm (10 in) long, its body was covered in bony scales.



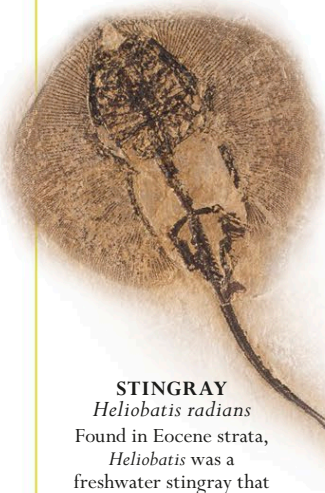
PSAMMOSTEID FISH

Drepanaspis sp.
A jawless, primitive fish, *Drepanaspis* had a flattened headshield. It is found only in Devonian strata.



PLACODERM

Bothriolepis canadensis
A Devonian placoderm (an extinct group of jawless fish), *Bothriolepis* had large head- and trunk-shields and spine-like pectoral fins.



STINGRAY

Heliobatis radians
Found in Eocene strata, *Heliobatis* was a freshwater stingray that grew to about 30 cm (12 in) in length and had a skeleton of cartilage.



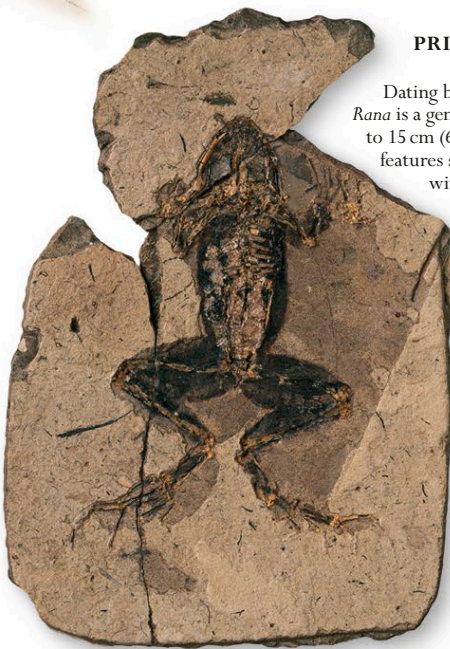
SHOAL OF DACE

Leuciscus pachecoi
Found in Miocene strata, extinct species of *Leuciscus* or dace resembled modern bony fish. *L. pachecoi* grew to 6 cm (2½ in) in length.



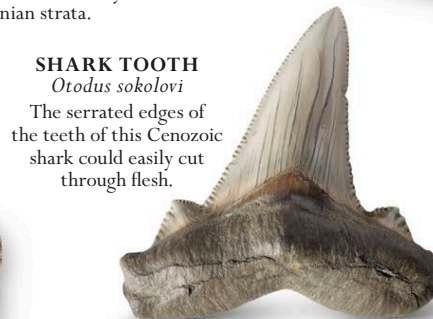
DIPLOCAULID AMPHIBIAN

Diplocaulus magnicornis
A salamander-like amphibian from the Permian Period, *Diplocaulus* had protrusions on the sides of its skull. It grew up to 1 m (3¼ ft) in length.



PRIMITIVE FROG

Rana pueyoi
Dating back to Miocene times, *Rana* is a genus of frogs. *R. pueyoi* grew to 15 cm (6 in) in length and shared features such as long hind limbs with modern frogs.



SHARK TOOTH

Otodus sokolovi
The serrated edges of the teeth of this Cenozoic shark could easily cut through flesh.

fine vertebral column

long, sharp tooth

SKULL OF LARGE, PREDATORY BONY FISH

Xiphactinus sp.
A bony fish from Late Cretaceous strata, *Xiphactinus* was a marine predator with a muscular body and large front teeth.

